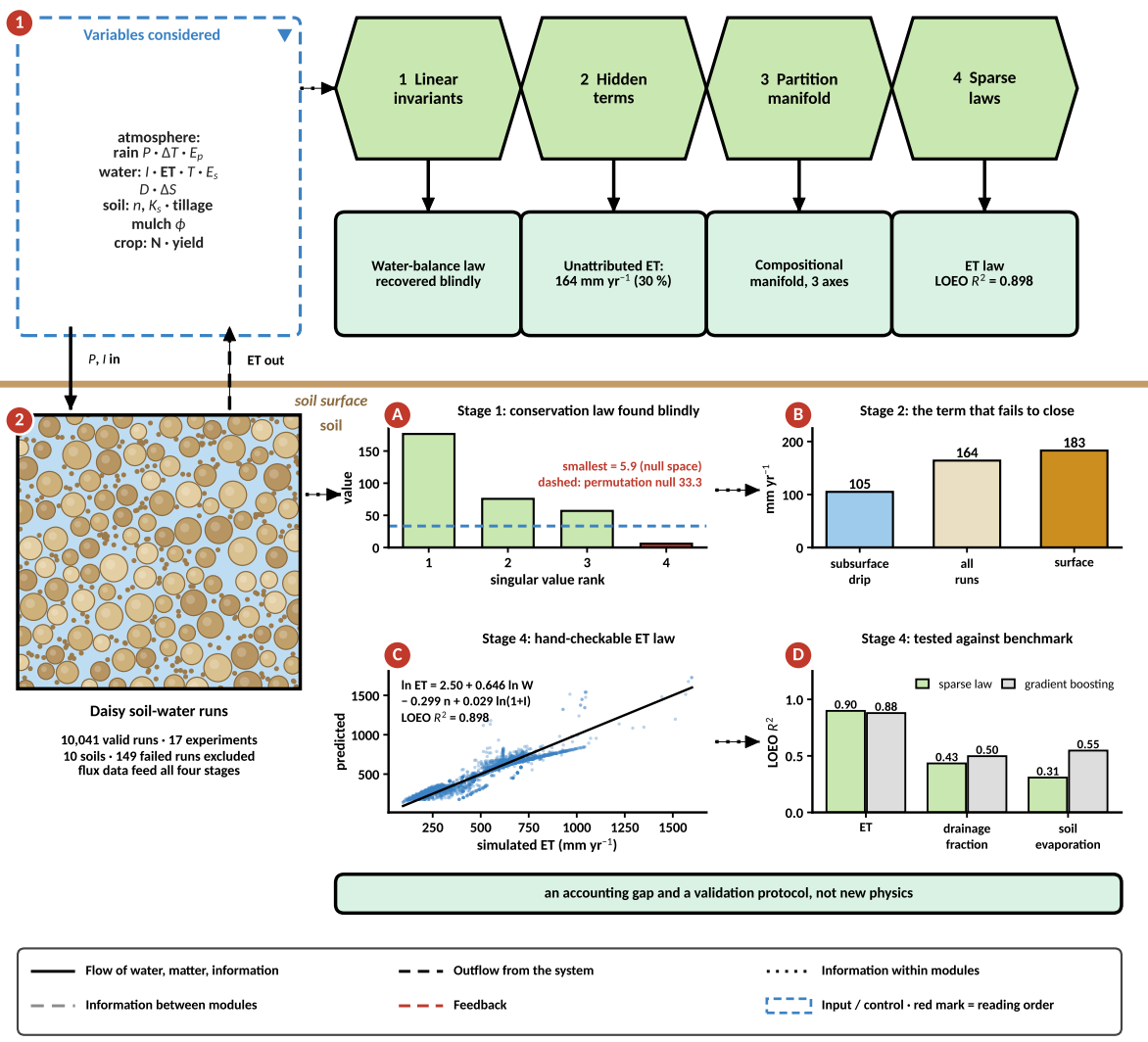

1
Variables considered
atmosphere:
rain P · ΔT · E_p
water: I · ET · T · E_s
D · ΔS
soil: n, K_s · tillage
mulch ϕ
crop: N · yield
1 Linear invariants
2 Hidden terms
3 Partition manifold
4 Sparse laws
Water-balance law recovered blindly
Unattributed ET: 164 mm yr^{-1} (30 %)
Compositional manifold, 3 axes
ET law LOEO R^2 = 0.898
P, I in
ET out
2
soil surface
soil
Daisy soil-water runs
10,041 valid runs · 17 experiments
10 soils · 149 failed runs excluded
flux data feed all four stages
A
Stage 1: conservation law found blindly
smallest = 5.9 (null space)
dashed: permutation null 33.3
value
singular value rank
B
Stage 2: the term that fails to close
mm yr^{-1}
105
164
183
subsurface drip
all runs
surface
C
Stage 4: hand-checkable ET law
ln ET = 2.50 + 0.646 ln W
− 0.299 n + 0.029 ln(1+I)
LOEO R^2 = 0.898
predicted
simulated ET (mm yr^{-1})
D
Stage 4: tested against benchmark
sparse law
gradient boosting
LOEO R^2
0.90
0.88
0.43
0.50
0.31
0.55
ET
drainage fraction
soil evaporation
an accounting gap and a validation protocol, not new physics
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Input / control · red mark = reading order